

Priya Balakrishnan (she/her/hers)

github.com/Bpriya42 | linkedin.com/in/priya-balakrishnan-sde | Website | bp42official@gmail.com | 413-512-3915

EDUCATION

University of Massachusetts Amherst

MS in Computer Science - Machine Learning and Artificial Intelligence (GPA: 3.83 / 4.0)

Amherst, MA, USA

Sept 2024 - May 2026

- Relevant Coursework: Neural Networks, Secure AI Systems, Natural Language Processing, Software Engineering.

University of Queensland

Bachelor of Engineering (Hons) in Electrical and Computer Engineering

Brisbane, Australia

Jul 2018 - Jul 2023

- Relevant Coursework: HCI, Machine Learning, Computer Architecture, OOPs in Java, Data Structures and Algorithms

EXPERIENCE

AI/ML Engineer Intern

New York City, NY, USA

Nevara AI

Jun 2025 - Aug 2025

- Built a full-stack AI recruiting platform (**React and TypeScript, Python, Fast API, Langchain and Supabase**), enabling recruiters to identify, train, and deliver high-quality candidates tailored to company needs.
- Designed and deployed a serverless voice assessment pipeline on AWS that processed 100% of candidate submissions and reduced manual review time by 90%, driving recruiter efficiency.
- Created an AI-powered evaluation engine by integrating Deepgram speech-to-text and OpenAI GPT-4o, analyzing voice features (pitch, fluency, filler words) into a 10-point rubric with actionable feedback.
- Designed and built 15+ responsive React/TypeScript pages powering 3 distinct assessment types (written, voice, and AI role-play assessments), delivering intuitive user flows that elevated candidate experience and recruiter adoption.
- Enhanced candidate-job matching by implementing geo-proximity and skill-based scoring, improving match accuracy to approximately 85% and strengthening recruiter trust in recommendations.

Full Stack Software Engineer

Brisbane, Australia

Deloitte

Jul 2023 - May 2024

- Implemented a secure, full-stack authentication system (**C#, ASP.NET, React/TypeScript, MySQL, REST APIs**) for Australia's largest FX provider, strengthening platform security for 100k+ customers.
- Built a real-time employee dashboard (**React, TypeScript, Vite**) to monitor exchange rates and manage customer data, improving operational visibility and decision-making.
- Integrated Auth0 identity management and MFA into the platform, reducing authentication-related support tickets by 60% and ensuring frictionless yet secure access.
- Automated migration of 10k+ user records with C# encryption/decryption, completing the process in less than 4 minutes and cutting manual effort by 90%.

Backend Software Engineer Intern

Brisbane, Australia

Deloitte

Nov 2021 - Dec 2021

- Built a multi-currency payment feature with **C#, REST APIs, ASP.NET, React (TypeScript), MySQL database** enabling real-time currency conversion and secure payment processing, improving transaction efficiency by 25%.

PROJECTS

AutoCareers – AI-Enhanced Job Search Platform

- Built and deployed a full-stack job search platform (**React, TypeScript, FastAPI, MongoDB, AWS**) reducing candidate job-hunt time by 45% with intelligent resume parsing, ATS scoring, and cover letter generation.
- Led frontend development of 12+ responsive pages, implemented CI/CD pipelines (Jest, Pytest, Postman), and automated API testing, increasing throughput by 30% and ensuring production-grade reliability.
- Architected a modular multi-agent AI system (OpenAI, TogetherAI, LlamaIndex) achieving 95% parsing accuracy and improving application quality across diverse resume formats.

Context Aggregation for Quality Automated Clinical Notes

- Architected an NLP pipeline to generate structured SOAP medical notes from clinician-patient dialogues, scaling execution on HPC clusters for concurrent experimentation.
- Engineered a multi-method classification system (embedding similarity, topic modeling, zero-shot transformers) achieving 91% accuracy and improving factual grounding by 15%.

PUBLICATION

Visualising Physiological Cues in Individualised Virtual Reality (iVR) for Mental Health

22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR - October 2023).

- Authored research paper and defined design principles for individualized VR visualizations for mental health treatment, under the guidance of Dr. Nilufar Baghaei, contributing to the field of Human Computer Interaction.
- Designed four VR visualizations for physiological data, developed expert and user testing protocols, and led usability study with 30+ participants, driving iterative design improvements to enhance User Experience.

SKILLS

Python | C# | Java | React | TypeScript | FastAPI | REST API | NoSQL | SQL | ASP.NET | CI/CD | Linux | Git | AWS | Docker | Kubernetes |